

reflection to calculate the distance using the simple formula $\text{distance} = (\text{speed} \times \text{time})/2$. Since the wave travels twice the distance due to reflection, we need to divide by '2' to get the actual distance.

- ◆ If the robot gets close to a wall while navigating (distance from wall directly in front is less than some value, say, 30 centimetres) rotate the ultrasonic sensor assembly to check distances on either sides to the robot.
- ◆ Steer the robot towards the direction where distance of the robot from the wall is the farthest.

This method is also used in the Micromouse event, where the distance of robots is measured from the wall. Then, an algorithm like flood fill helps the mouse travel to the centre and then calculates the route that the robot can travel in minimum time incase multiple routes exist.

Turn signalling biking jacket

This project would be especially useful if you or friends and family ride/cycle at night. It will help avoid accidents caused by motorists not being able to see you and knowing whether you're turning left or right.

To build a turn signalling jacket, you'll need the following components:

- ◆ LilyPad Arduino™ FTDI connector
- ◆ LED lights
- ◆ Two push button switches
- ◆ Mini USB cable
- ◆ LilyPad power supply
- ◆ Conductive thread

How to implement:

- ◆ Use the LilyPad Arduino™ to detect when the push button switch is pressed by the user.
- ◆ If the user presses the switch, change the state of LED connected to that side:
 - Turn OFF the LED, if LED are already ON.
 - Turn ON the LED, if they're already OFF.
 - You can use 'digitalWrite' function to achieve this functionality.
- ◆ Place the LilyPad Arduino™ on your jacket along with the batteries.
- ◆ Wire the LEFT turn LED together and the RIGHT turn LED together as shown in the picture.
- ◆ Connect the batteries and test the circuit.
- ◆ The FTDI connector will be required to program the LilyPad Arduino™.